

Green Hydrogen Plant Entrepreneur

Qualification Pack: SGJ/Q0121 | NSQF Level: 5 | Total Duration: 30 Hours (1,800 Mins)

Module 1: Introduction to Green Hydrogen (3.0 Hours)

UNIT 1.1 Necessity of Green Hydrogen in Sustainable Energy Transition (0.25 Hours / 15 Mins)

- 1.1.1 Introduction
- 1.1.2 Green Hydrogen in Sustainable Energy Transition

UNIT 1.2 Properties, Characteristics and Nomenclature of Hydrogen (0.50 Hours / 30 Mins)

- 1.2.1 Atomic Structure
- 1.2.2 Properties of Hydrogen
- 1.2.3 Hydrogen as an Energy Carrier
- 1.2.4 Colour Code Nomenclature of Hydrogen

UNIT 1.3 Value Chain for Green Hydrogen (0.25 Hours / 15 Mins)

- 1.3.1 Value Chain
- 1.3.2 Benefits and Drawbacks of Existing Methods of Hydrogen Production

UNIT 1.4 Key Aspects and Challenges related to Production, Storage, Transportation, and Distribution of Green Hydrogen (0.50 Hours / 30 Mins)

- 1.4.1 Challenges in Production of Green Hydrogen
- 1.4.2 Challenges in Storage of Green Hydrogen
- 1.4.3 Challenges in Transportation of Green Hydrogen
- 1.4.4 Challenges in the Distribution of Green Hydrogen

UNIT 1.5 End Use Applications of Green Hydrogen in Industry, Transport and Power Production (0.50 Hours / 30 Mins)

- 1.5.1 End Use Application of Green Hydrogen in Industry
- 1.5.2 End Use Application of Green Hydrogen in Transport Sector
- 1.5.3 End Use Application of Green Hydrogen in Power Sector

UNIT 1.6 Technology Options for Production of Green Hydrogen (0.25 Hours / 15 Mins)

- 1.6.1 Electrolysis Process
- 1.6.2 Types of Electrolysis

UNIT 1.7 Renewable Energy Sources for Large Scale Production of Green Hydrogen through Water Electrolysis (0.25 Hours / 15 Mins)

- 1.7.1 Renewable Energy Sources

UNIT 1.8 Emerging Entrepreneurial Opportunities Across Green Hydrogen Value Chain (0.25 Hours / 15 Mins)

- 1.8.1 Emerging Entrepreneurial Opportunities
- 1.8.2 Opportunities to Various Renewable Energy Sources

UNIT 1.9 Role and Responsibilities of a Green Hydrogen Plant Entrepreneur (0.25 Hours / 15 Mins)

- 1.9.1 Role and Responsibilities
- 1.9.2 Career Progression for Green Hydrogen Plant Entrepreneur

Module 2: Components of Green Hydrogen Plant and its Layout (6.0 Hours)

UNIT 2.1 Key Components of the Green Hydrogen Plant (1.00 Hour / 60 Mins)

- 2.1.1 Green Hydrogen Production Plant
- 2.1.2 Key Components and Their Categorisation
- 2.1.3 Mechanical Components
- 2.1.4 Civil Components

UNIT 2.2 Functions of Key Components and Their Fundamental Principles (0.75 Hours / 45 Mins)

- 2.2.1 Function of Electrical Components
- 2.2.2 Functions of Mechanical Components

2.2.3 Fundamental Principles of Main Components

UNIT 2.3 Types of Electrolyzers: Brief Description (0.75 Hours / 45 Mins)

2.3.1 Brief Description of Electrolyzers

UNIT 2.4 Overall Layout of the Plant and Sparking Items (0.75 Hours / 45 Mins)

2.4.1 Green Hydrogen Plant Layout

2.4.2 Optimum Utilization of the Energy and Materials

2.4.3 Sparking Items

UNIT 2.5 Read, Interpret and Understand the Relevant Drawings and O&M Manuals of the Plant Equipment (0.75 Hours / 45 Mins)

2.5.1 Engineering Drawings

2.5.2 O&M Manual

2.5.3 Codes and Standards for Referring in Drawings/Manuals

UNIT 2.6 Key Material and Safety Codes, Technology Protocols and Standards Applicable in Green Industry (1.50 Hours / 90 Mins)

2.6.1 Key Materials

2.6.2 Hydrogen Safety Challenges

2.6.3 Hydrogen Leak Detection

2.6.4 Indian Standards on Hydrogen

2.6.5 Personal Protective Equipment

2.6.6 Hazards and Safety Measures

2.6.7 Fire Prevention and Fire Fighting

2.6.8 Safe Distance from Sparking Devices

2.6.9 Training and Health Monitoring

UNIT 2.7 Key Technical Insights for Setting Up and Running Successful Enterprise (0.50 Hours / 30 Mins)

2.7.1 Key Technical Insights

Module 3: Key Technical and Entrepreneurial Aspects for Supporting Growth and Business Development in Green Hydrogen Production (6.0 Hours)

UNIT 3.1 Setting Up Green Hydrogen Production Plant (2.25 Hours / 135 Mins)

3.1.1 Setting up a Hydrogen Production Plant

3.1.2 Registrations Required for Setting Up Hydrogen Production Plant

3.1.3 Availing Low-Cost Financing

3.1.4 Evolving Opportunities

3.1.5 Key Ingredients of a Business Plan

3.1.6 Assess the Potential and Demand of Hydrogen in Different Sectors

UNIT 3.2 Cost Estimation (1.50 Hours / 90 Mins)

3.2.1 Lifecycle Cost

3.2.2 Key Aspects to Successfully Install, Operate, Safely Store and Market Hydrogen as Energy Source

3.2.3 O&M Related Cost for the System

UNIT 3.3 Taxation and Related Compliances (1.00 Hour / 60 Mins)

3.3.1 Taxation and Related Compliances in India

UNIT 3.4 Policies and Corresponding Incentives/Subsidies (1.25 Hours / 75 Mins)

3.4.1 National Green Hydrogen Policy

3.4.2 Evolving Start-Ups and New Ventures Boost-Up Hydrogen Economy

Module 4: Oversee the Assembly, Storage and O&M of Electrolyzer for Green Hydrogen Production (6.0 Hours)

UNIT 4.1 Types of Electrolyzers, their Technology Maturity and Technical Specifications (0.75 Hours / 45 Mins)

- 4.1.1 Types and Technological Status of Electrolyzers
- 4.1.2 Key Technical Specifications of Various Types of Electrolyzer
- 4.1.3 Comparing Various Electrolyzers
- 4.1.4 Operating Principles of PEM, AE and SOEC Electrolyzer

UNIT 4.2 Tools and Equipment for Installation (0.75 Hours / 45 Mins)

- 4.2.1 Installation of Electrolyzers
- 4.2.2 Technical Data Sheet

UNIT 4.3 Inputs/Outputs of an Electrolyzer System and Calculate Losses and Equipment Efficiency (0.75 Hours / 45 Mins)

- 4.3.1 Inputs/Outputs of an Electrolyzer System
- 4.3.2 Calculate Losses and Equipment Efficiency

UNIT 4.4 Input Renewable Power from Various Sources and its Integration with Electrolyzer (0.75 Hours / 45 Mins)

- 4.4.1 Renewable Sources of Energy and Their Limitations
- 4.4.2 Integration of Renewable Energy

UNIT 4.5 Assembly/Installation of Parts and Components of Electrolyzer (0.75 Hours / 45 Mins)

- 4.5.1 Assembly of Electrolyzer
- 4.5.2 PEM Electrolyzer/Fuel Cell Assembly

UNIT 4.6 Challenges Associated with Hydrogen in Storage, Handling and Transportation (0.75 Hours / 45 Mins)

- 4.6.1 Need for Hydrogen Storage
- 4.6.2 Challenges in Storing Hydrogen

UNIT 4.7 Hydrogen Compression System (0.75 Hours / 45 Mins)

- 4.7.1 Selection of Compression System
- 4.7.2 Precautions
- 4.7.3 Key Components/Parameters

UNIT 4.8 O&M Requirements for Electrolyzer (0.75 Hours / 45 Mins)

- 4.8.1 Operation and Maintenance (O&M) Activities
- 4.8.2 Safety Precautions

Module 5: Micro-Entrepreneurship Opportunities in Green Hydrogen (6.0 Hours)

UNIT 5.1 Micro-entrepreneurship Opportunities for Water Supply, Installation and Commissioning of Electrolyzer (1.25 Hours / 75 Mins)

- 5.1.1 Micro-Entrepreneurship Opportunities for Uninterrupted Quality Water Supply
- 5.1.2 Handling of Salt Enriched Water from Water Feed Input System
- 5.1.3 Installation and Commissioning of Electrolyzer

UNIT 5.2 Micro-entrepreneurship Opportunities for Renewable Energy Power Supply (1.00 Hour / 60 Mins)

- 5.2.1 Renewable Energy Power Supply Arrangement for Green Hydrogen Plant

UNIT 5.3 Micro-entrepreneurship Opportunities for Storage, Transportation, and Installation (1.25 Hours / 75 Mins)

- 5.3.1 Micro-Entrepreneurship Opportunities for Storage of Green Hydrogen
- 5.3.2 Micro-Entrepreneurship Opportunities for Safe Transportation of Green Hydrogen to Consumers
- 5.3.3 Micro-Entrepreneurship Opportunities for Installation of Pipelines in Green Hydrogen Plant/Industry

UNIT 5.4 Micro-entrepreneurship for Safety, Operation and Maintenance of Green Hydrogen Plant (1.25 Hours / 75 Mins)

- 5.4.1 Micro-entrepreneurship Opportunities for Managing Safety of the Plant
- 5.4.2 Micro-entrepreneurship Opportunities for Operation and Maintenance of Green Hydrogen Plant

UNIT 5.5 Micro-entrepreneurship for Leakage Monitoring (1.25 Hours / 75 Mins)

- 5.5.1 Micro-Entrepreneurship Opportunities for Leakage Monitoring
- 5.5.2 Micro-Entrepreneurship Opportunities for Purging of Hydrogen by CO₂/NH₃ Before Start of Work
- 5.5.3 Micro-Entrepreneurship Opportunities for Recharging of Hydrogen by Avoiding Air Contact

Module 6: Perform Health and Safety Measures for Installing and Operating Green Hydrogen Systems (3.0 Hours)

UNIT 6.1 Occupational Health & Safety Standards and Regulations (0.25 Hours / 15 Mins)

- 6.1.1 Introduction
- 6.1.2 Health Effects of Hydrogen
- 6.1.3 OSHA
- 6.1.4 Standards

UNIT 6.2 Emergency Situations in a Hydrogen System (0.25 Hours / 15 Mins)

- 6.2.1 Emergency Situations in a Hydrogen System

UNIT 6.3 Detectors and Safety Tools (0.25 Hours / 15 Mins)

- 6.3.1 Hydrogen Leak Detection
- 6.3.2 Methods of Hydrogen Leak Detection
- 6.3.3 Controls and Protection
- 6.3.4 Safety Tools

UNIT 6.4 Material Safety (0.25 Hours / 15 Mins)

- 6.4.1 Chemical Labels
- 6.4.2 Use of MSDS for Safe Handling

UNIT 6.5 Maintain Ideal Temperature and Humidity Level (0.25 Hours / 15 Mins)

- 6.5.1 Effect of Temperature and Humidity on Hydrogen Purity

UNIT 6.6 Smart Hazardous Chemical Storage Monitoring (0.25 Hours / 15 Mins)

- 6.6.1 Smart Monitoring

UNIT 6.7 Administering First Aid (0.25 Hours / 15 Mins)

- 6.7.1 Health Monitoring
- 6.7.2 First Aid and Medical Treatment

UNIT 6.8 Personal Protective Equipment (0.25 Hours / 15 Mins)

- 6.8.1 Personal Protective Equipment

UNIT 6.9 Hazards Associated with Hydrogen Production System (0.25 Hours / 15 Mins)

- 6.9.1 Hydrogen Production Equipment Related Hazards
- 6.9.2 Spillage and Leakage Hazards

UNIT 6.10 Work Safety Procedures and Instructions (0.25 Hours / 15 Mins)

- 6.10.1 Introduction
- 6.10.2 Safety Guidelines
- 6.10.3 Training
- 6.10.4 Storage, Handling, and Labelling
- 6.10.5 Transportation of Hydrogen

UNIT 6.11 Fire Protection (0.25 Hours / 15 Mins)

- 6.11.1 Introduction
- 6.11.2 Statutory Requirements for Fire Protection
- 6.11.3 Best Practices
- 6.11.4 Hydrogen Fire Fighting

UNIT 6.12 Housekeeping and Infection Control (0.25 Hours / 15 Mins)

- 6.12.1 Introduction
- 6.12.2 Housekeeping

6.12.3 Infection Control

6.12.4 Dos and Don'ts During Normal Operation and Maintenance

6.12.5 Dos and Don'ts During Event of an Accident